



University  
of Manitoba

# **Matrix methods – QR factorisation & SVD**

**MATH 2740 – Mathematics of Data Science – Unit U04**

**Julien Arino**

`julien.arino@umanitoba.ca`

**Department of Mathematics @ University of Manitoba**

**Fall 2026**

The University of Manitoba campuses are located on original lands of Anishinaabeg, Ininew, Anisininew, Dakota and Dene peoples, and on the National Homeland of the Red River Métis. We respect the Treaties that were made on these territories, we acknowledge the harms and mistakes of the past, and we dedicate ourselves to move forward in partnership with Indigenous communities in a spirit of Reconciliation and collaboration.

# Outline

**The QR decomposition**

**The SVD (singular values decomposition)**

# Matrix factorisations

Matrix factorisations are popular because they allow to perform some computations more easily

There are several different types of factorisations. Here, we study just the QR factorisation (useful for many least squares problems) and SVD (useful in many contexts)

**The QR decomposition**

**The SVD (singular values decomposition)**

# **The QR decomposition**

**Orthogonality and projections**

**The Gram-Schmidt orthogonalisation process**

**Orthogonal matrices**

**The QR factorisation & Least squares**

### Definition 57 (Orthogonal set of vectors)

The set of vectors  $\{\mathbf{v}_1, \dots, \mathbf{v}_k\} \in \mathbb{R}^n$  is an **orthogonal set** if

$$\forall i, j = 1, \dots, k, \quad i \neq j \implies \mathbf{v}_i \bullet \mathbf{v}_j = 0$$

### Theorem 58

$\{\mathbf{v}_1, \dots, \mathbf{v}_k\} \in \mathbb{R}^n$  with  $\forall i, \mathbf{v}_i \neq \mathbf{0}$ , orthogonal set  $\implies \{\mathbf{v}_1, \dots, \mathbf{v}_k\} \in \mathbb{R}^n$  linearly independent

### Definition 59 (Orthogonal basis)

Let  $S$  be a basis of the subspace  $W \subset \mathbb{R}^n$  composed of an orthogonal set of vectors. We say  $S$  is an **orthogonal basis** of  $W$

## Proof of Theorem 58

Assume  $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$  orthogonal set with  $\mathbf{v}_i \neq \mathbf{0}$  for all  $i = 1, \dots, k$ . Recall  $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$  is LI if

$$c_1 \mathbf{v}_1 + \dots + c_k \mathbf{v}_k = \mathbf{0} \iff c_1 = \dots = c_k = 0$$

So assume  $c_1, \dots, c_k \in \mathbb{R}$  are s.t.  $c_1 \mathbf{v}_1 + \dots + c_k \mathbf{v}_k = \mathbf{0}$ . Recall that  $\forall \mathbf{x} \in \mathbb{R}^k$ ,  $\mathbf{0}_k \bullet \mathbf{x} = 0$ . So for some  $\mathbf{v}_i \in \{\mathbf{v}_1, \dots, \mathbf{v}_k\}$

$$\begin{aligned} 0 &= \mathbf{0} \bullet \mathbf{v}_i \\ &= (c_1 \mathbf{v}_1 + \dots + c_k \mathbf{v}_k) \bullet \mathbf{v}_i \\ &= c_1 \mathbf{v}_1 \bullet \mathbf{v}_i + \dots + c_k \mathbf{v}_k \bullet \mathbf{v}_i \end{aligned} \tag{1}$$

As  $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$  orthogonal,  $\mathbf{v}_j \bullet \mathbf{v}_i = 0$  when  $i \neq j$ , (1) reduces to

$$c_i \mathbf{v}_i \bullet \mathbf{v}_i = 0 \iff c_i \|\mathbf{v}_i\|^2 = 0$$

As  $\mathbf{v}_i \neq \mathbf{0}$  for all  $i$ ,  $\|\mathbf{v}_i\| \neq 0$  and so  $c_i = 0$ . This is true for all  $i$ , hence the result □

## Example – Vectors of the standard basis of $\mathbb{R}^3$

For  $\mathbb{R}^3$ , we denote

$$\mathbf{i} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{j} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \text{ and } \mathbf{k} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

( $\mathbb{R}^k$  for  $k > 3$ , we denote them  $\mathbf{e}_i$ )

Clearly,  $\{\mathbf{i}, \mathbf{j}\}$ ,  $\{\mathbf{i}, \mathbf{k}\}$ ,  $\{\mathbf{j}, \mathbf{k}\}$  and  $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$  orthogonal sets. The standard basis vectors are also  $\neq \mathbf{0}$ , so the sets are LI. And

$$\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$$

is an orthogonal basis of  $\mathbb{R}^3$  since it spans  $\mathbb{R}^3$  and is LI

$$c_1 \mathbf{i} + c_2 \mathbf{j} + c_3 \mathbf{k} = c_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + c_3 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}$$

## Orthonormal version of things

### Definition 60 (Orthonormal set)

The set of vectors  $\{\mathbf{v}_1, \dots, \mathbf{v}_k\} \in \mathbb{R}^n$  is an **orthonormal set** if it is an orthogonal set and furthermore

$$\forall i = 1, \dots, k, \quad \|\mathbf{v}_i\| = 1$$

### Definition 61 (Orthonormal basis)

A basis of the subspace  $W \subset \mathbb{R}^n$  is an **orthonormal basis** if the vectors composing it are an orthonormal set

$\{\mathbf{v}_1, \dots, \mathbf{v}_k\} \in \mathbb{R}^n$  is orthonormal if

$$\mathbf{v}_i \bullet \mathbf{v}_j = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}$$

# **The QR decomposition**

Orthogonality and projections

**The Gram-Schmidt orthogonalisation process**

Orthogonal matrices

**The QR factorisation & Least squares**

What this aims to do is to construct an orthogonal basis for a subspace  $W \subset \mathbb{R}^n$

To do this, we use the *Gram-Schmidt orthogonalisation process*, which turns a basis of  $W$  into an orthogonal basis of  $W$

# Projections

## Definition 62 (Orthogonal projection onto a subspace)

$W \subset \mathbb{R}^n$  a subspace and  $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$  an orthogonal basis of  $W$ .  $\forall \mathbf{v} \in \mathbb{R}^n$ , the **orthogonal projection** of  $\mathbf{v}$  onto  $W$  is

$$\text{proj}_W(\mathbf{v}) = \frac{\mathbf{u}_1 \bullet \mathbf{v}}{\|\mathbf{u}_1\|^2} \mathbf{u}_1 + \dots + \frac{\mathbf{u}_k \bullet \mathbf{v}}{\|\mathbf{u}_k\|^2} \mathbf{u}_k$$

## Definition 63 (Component orthogonal to a subspace)

$W \subset \mathbb{R}^n$  a subspace and  $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$  an orthogonal basis of  $W$ .  $\forall \mathbf{v} \in \mathbb{R}^n$ , the **component** of  $\mathbf{v}$  orthogonal to  $W$  is

$$\text{perp}_W(\mathbf{v}) = \mathbf{v} - \text{proj}_W(\mathbf{v})$$

## Theorem 64 (Gram-Schmidt orthogonalisation process)

Let  $W$  be a subspace of  $\mathbb{R}^n$  and  $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$  be a basis of  $W$ . Let

$$\mathbf{v}_1 = \mathbf{x}_1$$

$$\mathbf{v}_2 = \mathbf{x}_2 - \frac{\mathbf{v}_1 \bullet \mathbf{x}_2}{\|\mathbf{v}_1\|^2} \mathbf{v}_1$$

$$\mathbf{v}_3 = \mathbf{x}_3 - \frac{\mathbf{v}_1 \bullet \mathbf{x}_3}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 - \frac{\mathbf{v}_2 \bullet \mathbf{x}_3}{\|\mathbf{v}_2\|^2} \mathbf{v}_2$$

$\vdots$

$$\mathbf{v}_k = \mathbf{x}_k - \frac{\mathbf{v}_1 \bullet \mathbf{x}_k}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 - \dots - \frac{\mathbf{v}_{k-1} \bullet \mathbf{x}_k}{\|\mathbf{v}_{k-1}\|^2} \mathbf{v}_{k-1}$$

and

$$W_1 = \text{span}(\mathbf{x}_1), W_2 = \text{span}(\mathbf{x}_1, \mathbf{x}_2), \dots, W_k = \text{span}(\mathbf{x}_1, \dots, \mathbf{x}_k)$$

Then  $\forall i = 1, \dots, k$ ,  $\{\mathbf{v}_1, \dots, \mathbf{v}_i\}$  is an orthogonal basis of  $W_i$

## Theorem 65 (Gram-Schmidt orthonormalisation process)

Let  $W$  be a subspace of  $\mathbb{R}^n$  and  $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$  be a basis of  $W$ . Let

$$\tilde{\mathbf{v}}_1 = \mathbf{x}_1 / \|\mathbf{x}_1\|$$

$$\mathbf{v}_2 = \mathbf{x}_2 - (\tilde{\mathbf{v}}_1 \bullet \mathbf{x}_2) \tilde{\mathbf{v}}_1$$

$$\tilde{\mathbf{v}}_2 = \frac{\mathbf{v}_2}{\|\mathbf{v}_2\|}$$

$$\mathbf{v}_3 = \mathbf{x}_3 - (\tilde{\mathbf{v}}_1 \bullet \mathbf{x}_3) \tilde{\mathbf{v}}_1 - (\tilde{\mathbf{v}}_2 \bullet \mathbf{x}_3) \tilde{\mathbf{v}}_2$$

$$\tilde{\mathbf{v}}_3 = \frac{\mathbf{v}_3}{\|\mathbf{v}_3\|}$$

$\vdots$

$$\mathbf{v}_k = \mathbf{x}_k - (\tilde{\mathbf{v}}_1 \bullet \mathbf{x}_k) \tilde{\mathbf{v}}_1 - \dots - (\tilde{\mathbf{v}}_{k-1} \bullet \mathbf{x}_k) \tilde{\mathbf{v}}_{k-1}$$

$$\tilde{\mathbf{v}}_k = \frac{\mathbf{v}_k}{\|\mathbf{v}_k\|}$$

and

$$W_1 = \text{span}(\mathbf{x}_1), W_2 = \text{span}(\mathbf{x}_1, \mathbf{x}_2), \dots, W_k = \text{span}(\mathbf{x}_1, \dots, \mathbf{x}_k)$$

Then  $\forall i = 1, \dots, k$ ,  $\{\tilde{\mathbf{v}}_1, \dots, \tilde{\mathbf{v}}_i\}$  is an orthonormal basis of  $W_i$

## Links between the two processes

- ▶ To get an orthonormal basis, you can also use the orthogonalisation process then normalise the result
- ▶ There is no difference between the two methods, the only gain in the orthonormalisation process is that you don't need to divide by  $\|\cdot\|^2$  when projecting, since you have already done so just before

# **The QR decomposition**

Orthogonality and projections

The Gram-Schmidt orthogonalisation process

**Orthogonal matrices**

The QR factorisation & Least squares

## Theorem 66

Let  $Q \in \mathcal{M}_{mn}$ . The columns of  $Q$  form an orthonormal set if and only if

$$Q^T Q = \mathbb{I}_n$$

## Definition 67 (Orthogonal matrix)

$Q \in \mathcal{M}_n$  is an **orthogonal matrix** if its columns form an orthonormal set

So  $Q \in \mathcal{M}_n$  orthogonal if  $Q^T Q = \mathbb{I}$ , i.e.,  $Q^T = Q^{-1}$

## Theorem 68 (NSC for orthogonality)

$$Q \in \mathcal{M}_n \text{ orthogonal} \iff Q^{-1} = Q^T$$

# Orthogonal matrices “encode” isometries

## Theorem 69 (Orthogonal matrices “encode” isometries)

Let  $Q \in \mathcal{M}_n$ . TFAE

1.  $Q$  orthogonal
2.  $\forall \mathbf{x} \in \mathbb{R}^n, \|Q\mathbf{x}\| = \|\mathbf{x}\|$
3.  $\forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n, Q\mathbf{x} \bullet Q\mathbf{y} = \mathbf{x} \bullet \mathbf{y}$

## Proof of Theorem 69

**1  $\implies$  3:** Assume  $Q$  is orthogonal. Then  $Q^T Q = \mathbb{I}_n$ , so

$$Q\mathbf{x} \bullet Q\mathbf{y} = (Q\mathbf{x})^T (Q\mathbf{y}) = \mathbf{x}^T Q^T Q \mathbf{y} = \mathbf{x}^T \mathbb{I}_n \mathbf{y} = \mathbf{x} \bullet \mathbf{y}$$

**3  $\implies$  1:** Assume  $Q\mathbf{x} \bullet Q\mathbf{y} = \mathbf{x} \bullet \mathbf{y} \forall \mathbf{x}, \mathbf{y}$ . For standard basis vectors  $\mathbf{e}_i, \mathbf{e}_j$ :

$$(Q\mathbf{e}_i) \bullet (Q\mathbf{e}_j) = \mathbf{e}_i \bullet \mathbf{e}_j = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

Since  $Q\mathbf{e}_i$  is the  $i$ -th column of  $Q$ , the columns of  $Q$  are orthonormal, so  $Q$  is orthogonal.

**3  $\implies$  2:** Assume  $Q\mathbf{x} \bullet Q\mathbf{y} = \mathbf{x} \bullet \mathbf{y}$ . Letting  $\mathbf{y} = \mathbf{x}$  gives

$$\|Q\mathbf{x}\|^2 = Q\mathbf{x} \bullet Q\mathbf{x} = \mathbf{x} \bullet \mathbf{x} = \|\mathbf{x}\|^2 \implies \|Q\mathbf{x}\| = \|\mathbf{x}\|$$

**2  $\implies$  3:** Assume  $\|Q\mathbf{z}\| = \|\mathbf{z}\|$  for all  $\mathbf{z}$ . Using  $\mathbf{z} = \mathbf{x} + \mathbf{y}$ , we expand  $\|Q(\mathbf{x} + \mathbf{y})\|^2 = \|\mathbf{x} + \mathbf{y}\|^2$ :

$$\|Q\mathbf{x}\|^2 + 2Q\mathbf{x} \bullet Q\mathbf{y} + \|Q\mathbf{y}\|^2 = \|\mathbf{x}\|^2 + 2\mathbf{x} \bullet \mathbf{y} + \|\mathbf{y}\|^2$$

Since  $\|Q\mathbf{x}\| = \|\mathbf{x}\|$  and  $\|Q\mathbf{y}\| = \|\mathbf{y}\|$ , we get  $Q\mathbf{x} \bullet Q\mathbf{y} = \mathbf{x} \bullet \mathbf{y}$ .



## Theorem 70

*Let  $Q \in \mathcal{M}_n$  be orthogonal. Then*

- 1. The rows of  $Q$  form an orthonormal set*
- 2.  $Q^{-1}$  orthogonal*
- 3.  $\det Q = \pm 1$*
- 4.  $\forall \lambda \in \sigma(Q), |\lambda| = 1$*
- 5. If  $Q_2 \in \mathcal{M}_n$  also orthogonal, then  $QQ_2$  orthogonal*

# Proof of 1 and 2 in Theorem 70

## 1. The rows of $Q$ form an orthonormal set

- ▶ By definition,  $Q$  is orthogonal  $\iff Q^T Q = \mathbb{I}_n$
- ▶ Since  $Q$  is square,  $Q^T Q = \mathbb{I}_n \implies Q^{-1} = Q^T \implies Q Q^T = \mathbb{I}_n$
- ▶ This is equivalent to  $(Q^T)^T Q^T = \mathbb{I}_n$ , so the columns of  $Q^T$  (the rows of  $Q$ ) are orthonormal □

## 2. $Q^{-1}$ is orthogonal

- ▶ We must show that  $(Q^{-1})^T Q^{-1} = \mathbb{I}_n$
- ▶ Using  $Q^{-1} = Q^T$ , we get  $(Q^{-1})^T Q^{-1} = (Q^T)^T Q^T = Q Q^T = \mathbb{I}_n$
- ▶ Thus  $Q^{-1}$  is an orthogonal matrix □

## Proof of 3 and 5 in Theorem 70

### 3. $\det Q = \pm 1$

- ▶ We have  $Q^T Q = \mathbb{I}_n$ . Taking determinant on both sides

$$\det(Q^T Q) = \det(\mathbb{I}_n) = 1$$

- ▶ Using  $\det(AB) = \det(A)\det(B)$ , we get  $\det(Q^T)\det(Q) = 1$
- ▶ From the general property  $\det(A^T) = \det(A)$ , it follows that  $(\det(Q))^2 = 1$ , so  $\det Q = \pm 1$  □

### 5. If $Q_2 \in \mathcal{M}_n$ also orthogonal, then $QQ_2$ orthogonal

- ▶ We must show that  $(QQ_2)^T(QQ_2) = \mathbb{I}_n$
- ▶  $(QQ_2)^T(QQ_2) = Q_2^T Q^T QQ_2 = Q_2^T (Q^T Q) Q_2$
- ▶ Since  $Q$  is orthogonal,  $Q^T Q = \mathbb{I}_n$ , so the expression becomes  $Q_2^T \mathbb{I}_n Q_2 = Q_2^T Q_2$
- ▶ Since  $Q_2$  is orthogonal,  $Q_2^T Q_2 = \mathbb{I}_n$  □

## Proof of 4 in Theorem 70

4.  $\forall \lambda \in \sigma(Q), |\lambda| = 1$

Let  $\lambda$  be an eigenvalue of  $Q \in \mathcal{M}_n$  orthogonal, i.e.,  $\exists \mathbb{R}^n \ni \mathbf{x} \neq \mathbf{0}$  s.t.

$$Q\mathbf{x} = \lambda\mathbf{x}$$

Take the norm on both sides

$$\|Q\mathbf{x}\| = \|\lambda\mathbf{x}\|$$

From 2 in Theorem 69,  $\|Q\mathbf{x}\| = \|\mathbf{x}\|$  and from the properties of norms,  $\|\lambda\mathbf{x}\| = |\lambda| \|\mathbf{x}\|$ , so we have

$$\|Q\mathbf{x}\| = \|\lambda\mathbf{x}\| \iff \|\mathbf{x}\| = |\lambda| \|\mathbf{x}\| \iff 1 = |\lambda|$$

(we can divide by  $\|\mathbf{x}\|$  since  $\mathbf{x} \neq \mathbf{0}$  as an eigenvector)



# The QR decomposition

Orthogonality and projections

The Gram-Schmidt orthogonalisation process

Orthogonal matrices

**The QR factorisation & Least squares**

# The QR factorisation

## Theorem 71

*Let  $A \in \mathcal{M}_{mn}$  with LI columns. Then  $A$  can be factored as*

$$A = QR$$

*where  $Q \in \mathcal{M}_{mn}$  has orthonormal columns and  $R \in \mathcal{M}_n$  is nonsingular upper triangular*

## Back to least squares

So what was the point of all that..?

Theorem 72 (Least squares with QR factorisation)

*$A \in \mathcal{M}_{mn}$  with linearly independent columns,  $\mathbf{b} \in \mathbb{R}^m$ . If  $A = QR$  is a QR factorisation of  $A$ , then the unique least squares solution  $\tilde{\mathbf{x}}$  of  $A\mathbf{x} = \mathbf{b}$  is*

$$\tilde{\mathbf{x}} = R^{-1}Q^T\mathbf{b}$$

## Proof of Theorem 72

$A$  has LI columns so

- ▶ least squares  $A\mathbf{x} = \mathbf{b}$  has unique solution  $\tilde{\mathbf{x}} = (A^T A)^{-1} A^T \mathbf{b}$
- ▶ by Theorem 71,  $A$  can be written as  $A = QR$  with  $Q \in \mathcal{M}_{mn}$  with orthonormal columns and  $R \in \mathcal{M}_n$  nonsingular and upper triangular

So

$$\begin{aligned} A^T A \tilde{\mathbf{x}} &= A^T \mathbf{b} \implies (QR)^T QR \tilde{\mathbf{x}} = (QR)^T \mathbf{b} \\ &\implies R^T Q^T QR \tilde{\mathbf{x}} = R^T Q^T \mathbf{b} \\ &\implies R^T \mathbb{I}_n R \tilde{\mathbf{x}} = R^T Q^T \mathbf{b} \\ &\implies R^T R \tilde{\mathbf{x}} = R^T Q^T \mathbf{b} \\ &\implies (R^T)^{-1} R \tilde{\mathbf{x}} = (R^T)^{-1} R^T Q^T \mathbf{b} \\ &\implies R \tilde{\mathbf{x}} = Q^T \mathbf{b} \\ &\implies \tilde{\mathbf{x}} = R^{-1} Q^T \mathbf{b} \quad \square \end{aligned}$$

## Other applications of the QR decomposition

Focus here on least squares, but QR decomposition fundamental tool in numerical linear algebra with other important applications

- ▶ **Solving linear systems:** to solve  $A\mathbf{x} = \mathbf{b}$  for a non-singular square matrix  $A$ , write  $QR\mathbf{x} = \mathbf{b} \implies R\mathbf{x} = Q^T\mathbf{b}$ . Since  $R$  is upper triangular, this can be quickly solved by back-substitution. This method is often more numerically stable than Gaussian elimination (called the LU decomposition)
- ▶ **Computing eigenvalues:** the **QR algorithm** is a standard method for calculating the eigenvalues of a matrix. It uses the iterative process  $A_k = Q_k R_k$  and  $A_{k+1} = R_k Q_k$ . Under certain conditions,  $A_k$  converges to an upper triangular matrix whose diagonal entries are the eigenvalues
- ▶ **Computing determinants:** if  $A = QR$ , then  $\det(A) = \det(Q)\det(R)$ . Since  $Q$  is orthogonal,  $\det(Q) = \pm 1$ , and since  $R$  is upper triangular,  $\det(R)$  is simply the product of its diagonal entries

## Example – Solving least squares with QR (1/5)

Consider the matrix  $A$  and a vector  $\mathbf{b}$ :

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad \mathbf{b} = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}.$$

We want to find the unique least squares solution  $\tilde{\mathbf{x}}$  to  $A\mathbf{x} = \mathbf{b}$ .

To do this using the QR decomposition, we must first compute  $Q$  and  $R$ .

## Example – Solving least squares with QR (2/5)

We compute  $Q$  by applying the Gram-Schmidt process to the columns of  $A$ :

$\mathbf{a}_1 = (1, 1, 0)$  and  $\mathbf{a}_2 = (1, 0, 1)$ .

First, we find the orthogonal vectors:

$$\mathbf{v}_1 = \mathbf{a}_1 = (1, 1, 0).$$

$$\mathbf{v}_2 = \mathbf{a}_2 - \frac{\mathbf{v}_1 \bullet \mathbf{a}_2}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 = (1, 0, 1) - \frac{1}{2}(1, 1, 0) = \left(\frac{1}{2}, -\frac{1}{2}, 1\right).$$

Then, we normalise them to obtain the orthonormal columns of  $Q$ :

$$\mathbf{u}_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right).$$

$$\mathbf{u}_2 = \frac{\mathbf{v}_2}{\|\mathbf{v}_2\|} = \sqrt{\frac{2}{3}} \left(\frac{1}{2}, -\frac{1}{2}, 1\right) = \left(\frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}\right).$$

## Example – Solving least squares with QR (3/5)

Thus, the orthogonal matrix  $Q$  is:

$$Q = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} \\ 0 & \frac{2}{\sqrt{6}} \end{pmatrix}.$$

To find  $R$ , recall that  $A = QR$ . Since  $Q$  has orthonormal columns,  $Q^T Q = \mathbb{I}$ . Thus:

$$Q^T A = Q^T QR = \mathbb{I}R = R.$$

We can therefore compute  $R$  directly as  $Q^T A$ :

$$R = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \frac{2}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & \frac{3}{\sqrt{6}} \end{pmatrix}.$$

## Example – Solving least squares with QR (4/5)

By Theorem 72, we know that  $R\tilde{\mathbf{x}} = Q^T \mathbf{b}$ . First, we compute  $Q^T \mathbf{b}$ :

$$Q^T \mathbf{b} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} = \begin{pmatrix} \frac{5}{\sqrt{2}} \\ \frac{7}{\sqrt{6}} \end{pmatrix}.$$

Now we solve the upper triangular system  $R\tilde{\mathbf{x}} = Q^T \mathbf{b}$  via back-substitution:

$$\begin{pmatrix} \frac{2}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & \frac{3}{\sqrt{6}} \end{pmatrix} \begin{pmatrix} \tilde{x}_1 \\ \tilde{x}_2 \end{pmatrix} = \begin{pmatrix} \frac{5}{\sqrt{2}} \\ \frac{7}{\sqrt{6}} \end{pmatrix}.$$

The second equation gives  $\frac{3}{\sqrt{6}} \tilde{x}_2 = \frac{7}{\sqrt{6}}$ , which implies  $\tilde{x}_2 = \frac{7}{3}$ .

## Example – Solving least squares with QR (5/5)

Substituting this into the first equation yields  $\frac{2}{\sqrt{2}}\tilde{x}_1 + \frac{1}{\sqrt{2}}\left(\frac{7}{3}\right) = \frac{5}{\sqrt{2}}$ , which simplifies to  $2\tilde{x}_1 + \frac{7}{3} = 5$ . Solving for  $\tilde{x}_1$  yields  $\tilde{x}_1 = \frac{4}{3}$ .

Thus, the least squares solution is  $\tilde{\mathbf{x}} = \begin{pmatrix} 4/3 \\ 7/3 \end{pmatrix}$ .

We can easily confirm this result in R:

```
b <- c(2, 3, 4)
QT_b <- t(Q_A) %*% b
x_tilde <- solve(R_A, QT_b)
x_tilde

##           [,1]
## [1,] 1.333333
## [2,] 2.333333

qr.coef(qr_A, b) # with the built-in solver

## [1] 1.333333 2.333333
```

## Fitting the Canada census data using QR

Let us return to the Canada census data we fitted earlier using the normal equations.

We saw that computing  $A^T A$  and its inverse for a quadratic fit was problematic when using the raw years, because the matrix  $A$  was badly conditioned.

This led us to shift the  $x$ -axis to start at  $t = 0$ .

The QR decomposition provides a much more numerically stable way to solve the least squares problem  $A\mathbf{x} = \mathbf{b}$ .

To illustrate this, let us try to fit the quadratic model on the unshifted raw data using the QR approach.

## Setup for unshifted quadratic fit

We first load the data and set up our matrix  $A$  for a quadratic fit without shifting the years.

```
data <- read.csv("../DATA/Canada_census.csv")
data <- data[which(data$year <= 1980),]
x <- data$year
y <- data$population
A_q <- matrix(c(rep(1, length(x)), x, x^2), nr = length(x), nc = 3)
```

## Normal equations failure

Recall that using the normal equations on this matrix failed because  $A^T A$  is computationally singular. We can see this in R:

```
ATA <- t(A_q) %*% A_q
try(solve(ATA) %*% t(A_q) %*% y)

## Error in solve.default(ATA) :
##   system is computationally singular: reciprocal condition number = 9.166
```

However, the QR decomposition avoids forming  $A^T A$ , preserving numerical stability.

## Solving with QR decomposition

We can compute the QR decomposition using R's built-in `qr` function, extract the  $Q$  and  $R$  components, and then use `backsolve` to solve the upper triangular system  $R\mathbf{x} = Q^T \mathbf{b}$ .

```
# Compute the QR decomposition of A_q
QR <- qr(A_q)
Q <- qr.Q(QR)
R <- qr.R(QR)

# Solve Rx = Q^T y
sol_qr <- backsolve(R, t(Q) %*% y)
sol_qr

##           [,1]
## [1,]  5.133645e+09
## [2,] -5.513814e+06
## [3,]  1.481439e+03
```

## Direct solution with `qr.solve`

The QR decomposition successfully finds the coefficients for the unshifted data!

Alternatively, R provides the function `qr.solve` which directly solves the least squares problem using the QR decomposition under the hood:

```
sol_qr_direct <- qr.solve(A_q, y)
sol_qr_direct

## [1] 5.133645e+09 -5.513814e+06 1.481439e+03
```

This demonstrates the practical advantage of using orthogonal matrix factorisations in data science applications over the normal equations.

**The QR decomposition**

**The SVD (singular values decomposition)**

# **The SVD (singular values decomposition)**

**Singular values**

**The SVD**

**An application of the SVD – Image compression**

**Computing the SVD**

**Applications of the SVD – Least squares**

**Summary – Least squares methods**

## Matrix factorisations (continued)

The singular value decomposition (known mostly by its acronym, SVD) is yet another type of factorisation/decomposition..

# Singular values

## Definition 73 (Singular value)

Let  $A \in \mathcal{M}_{mn}(\mathbb{R})$ . The **singular values** of  $A$  are the real numbers

$$\sigma_1 \geq \sigma_2 \geq \cdots \sigma_n \geq 0$$

that are the square roots of the eigenvalues of  $A^T A$

## Singular values are real and nonnegative?

Recall that  $\forall A \in \mathcal{M}_{mn}$ ,  $A^T A$  is symmetric

**Claim 1.** Real symmetric matrices have real eigenvalues

**Proof.**  $A \in \mathcal{M}_n(\mathbb{R})$  symmetric and  $(\lambda, \mathbf{v})$  eigenpair of  $A$ , i.e,  $A\mathbf{v} = \lambda\mathbf{v}$ . Taking the complex conjugate,  $\overline{A\mathbf{v}} = \overline{\lambda\mathbf{v}}$

Since  $A \in \mathcal{M}_n(\mathbb{R})$ ,  $\overline{A} = A$  ( $z = \bar{z} \iff z \in \mathbb{R}$ )

So

$$A\bar{\mathbf{v}} = \overline{A\mathbf{v}} = \overline{\lambda\mathbf{v}} = \overline{\lambda}\bar{\mathbf{v}} = \bar{\lambda}\bar{\mathbf{v}}$$

i.e., if  $(\lambda, \mathbf{v})$  eigenpair,  $(\bar{\lambda}, \bar{\mathbf{v}})$  also eigenpair

Still assuming  $A \in \mathcal{M}_n(\mathbb{R})$  symmetric and  $(\lambda, \mathbf{v})$  eigenpair of  $A$  and using what we just proved (that  $(\bar{\lambda}, \bar{\mathbf{v}})$  also eigenpair), take transposes

$$\begin{aligned} A\bar{\mathbf{v}} = \bar{\lambda}\bar{\mathbf{v}} &\iff (A\bar{\mathbf{v}})^T = (\bar{\lambda}\bar{\mathbf{v}})^T \\ &\iff \bar{\mathbf{v}}^T A^T = \bar{\lambda}\bar{\mathbf{v}}^T \\ &\iff \bar{\mathbf{v}}^T A = \bar{\lambda}\bar{\mathbf{v}}^T \quad [A \text{ symmetric}] \end{aligned}$$

Let us now compute  $\lambda(\bar{\mathbf{v}} \bullet \mathbf{v})$ . We have

$$\begin{aligned} \lambda(\bar{\mathbf{v}} \bullet \mathbf{v}) &= \lambda \bar{\mathbf{v}}^T \mathbf{v} = \bar{\mathbf{v}}^T (\lambda \mathbf{v}) \\ &= \bar{\mathbf{v}}^T (A\mathbf{v}) = (\bar{\mathbf{v}}^T A) \mathbf{v} \\ &= (\bar{\lambda} \bar{\mathbf{v}}^T) \mathbf{v} = \bar{\lambda}(\bar{\mathbf{v}} \bullet \mathbf{v}) \\ &\iff (\lambda - \bar{\lambda})(\bar{\mathbf{v}} \bullet \mathbf{v}) = 0 \end{aligned}$$

We have shown

$$(\lambda - \bar{\lambda})(\bar{\mathbf{v}} \bullet \mathbf{v}) = 0$$

Let

$$\mathbf{v} = \begin{pmatrix} a_1 + ib_1 \\ \vdots \\ a_n + ib_n \end{pmatrix}$$

Then

$$\bar{\mathbf{v}} = \begin{pmatrix} a_1 - ib_1 \\ \vdots \\ a_n - ib_n \end{pmatrix}$$

So

$$\bar{\mathbf{v}} \bullet \mathbf{v} = (a_1^2 + b_1^2) + \cdots + (a_n^2 + b_n^2)$$

But  $\mathbf{v}$  eigenvector is  $\neq \mathbf{0}$ , so  $\bar{\mathbf{v}} \bullet \mathbf{v} \neq 0$ , so

$$(\lambda - \bar{\lambda})(\bar{\mathbf{v}} \bullet \mathbf{v}) = 0 \iff \lambda - \bar{\lambda} = 0 \iff \lambda = \bar{\lambda} \iff \lambda \in \mathbb{R} \quad \square$$

**Claim 2.** For  $A \in \mathcal{M}_{mn}(\mathbb{R})$ , the eigenvalues of  $A^T A$  are real and nonnegative

**Proof.** We know that for  $A \in \mathcal{M}_{mn}$ ,  $A^T A$  symmetric and from previous claim, if  $A \in \mathcal{M}_{mn}(\mathbb{R})$ , then  $A^T A$  is symmetric and real and with real eigenvalues

Let  $(\lambda, \mathbf{v})$  be an eigenpair of  $A^T A$ , with  $\mathbf{v}$  chosen so that  $\|\mathbf{v}\| = 1$

Norms are functions  $V \rightarrow \mathbb{R}_+$ , so  $\|A\mathbf{v}\|$  and  $\|A\mathbf{v}\|^2$  are  $\geq 0$  and thus

$$\begin{aligned} 0 \leq \|A\mathbf{v}\|^2 &= (A\mathbf{v}) \bullet (A\mathbf{v}) = (A\mathbf{v})^T (A\mathbf{v}) \\ &= \mathbf{v}^T A^T A \mathbf{v} = \mathbf{v}^T (A^T A \mathbf{v}) = \mathbf{v}^T (\lambda \mathbf{v}) \\ &= \lambda (\mathbf{v}^T \mathbf{v}) = \lambda (\mathbf{v} \bullet \mathbf{v}) = \lambda \|\mathbf{v}\|^2 \\ &= \lambda \end{aligned}$$

**Claim 3.** For  $A \in \mathcal{M}_{mn}(\mathbb{R})$ , the nonzero eigenvalues of  $A^T A$  and  $AA^T$  are the same

**Proof.** Let  $(\lambda, \mathbf{v})$  be an eigenpair of  $A^T A$  with  $\lambda \neq 0$ . Then  $\mathbf{v} \neq \mathbf{0}$  and

$$A^T A \mathbf{v} = \lambda \mathbf{v} \neq \mathbf{0}$$

Left multiply by  $A$

$$AA^T A \mathbf{v} = \lambda A \mathbf{v}$$

Let  $\mathbf{w} = A \mathbf{v}$ , we thus have  $AA^T \mathbf{w} = \lambda \mathbf{w}$ ; in other words,  $A \mathbf{v}$  is an eigenvector of  $AA^T$  corresponding to the (nonzero) eigenvalue  $\lambda$

The reverse works the same way..



# **The SVD (singular values decomposition)**

Singular values

**The SVD**

An application of the SVD – Image compression

Computing the SVD

Applications of the SVD – Least squares

Summary – Least squares methods

# The singular value decomposition (SVD)

## Theorem 74 (SVD)

*$A \in \mathcal{M}_{mn}$  with singular values  $\sigma_1 \geq \dots \geq \sigma_r > 0$  and  $\sigma_{r+1} = \dots = \sigma_n = 0$*

*Then there exists  $U \in \mathcal{M}_m$  orthogonal,  $V \in \mathcal{M}_n$  orthogonal and a block matrix  $\Sigma \in \mathcal{M}_{mn}$  taking the form*

$$\Sigma = \begin{pmatrix} D & 0_{r,n-r} \\ 0_{m-r,r} & 0_{m-r,n-r} \end{pmatrix}$$

*where*

$$D = \text{diag}(\sigma_1, \dots, \sigma_r) \in \mathcal{M}_r$$

*such that*

$$A = U\Sigma V^T$$

## Definition 75

We call a factorisation as in Theorem 74 the **singular value decomposition** of  $A$ . The columns of  $U$  and  $V$  are, respectively, the **left** and **right singular vectors** of  $A$

$U$  and  $V^T$  are *rotation* or *reflection* matrices,  $\Sigma$  is a *scaling* matrix

$U \in \mathcal{M}_m$  orthogonal matrix with columns the eigenvectors of  $AA^T$

$V \in \mathcal{M}_n$  orthogonal matrix with columns the eigenvectors of  $A^T A$

# Outer product form of the SVD

## Theorem 76 (Outer product form of the SVD)

*$A \in \mathcal{M}_{mn}$  with singular values  $\sigma_1 \geq \dots \geq \sigma_r > 0$  and  $\sigma_{r+1} = \dots = \sigma_n = 0$ ,  $\mathbf{u}_1, \dots, \mathbf{u}_r$  and  $\mathbf{v}_1, \dots, \mathbf{v}_r$ , respectively, left and right singular vectors of  $A$  corresponding to these singular values*

*Then*

$$A = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \dots + \sigma_r \mathbf{u}_r \mathbf{v}_r^T \quad (2)$$

# **The SVD (singular values decomposition)**

Singular values

The SVD

**An application of the SVD – Image compression**

Computing the SVD

Applications of the SVD – Least squares

Summary – Least squares methods

# Applications of the SVD

Many applications of the SVD, both theoretical and practical..

1. Obtaining a unique solutions to least squares when  $A^T A$  singular
2. Image compression

# Compressing images

Consider an image (for simplicity, assume in shades of grey). This can be stored in a matrix  $A \in \mathcal{M}_{mn}$

Take the SVD of  $A$ . Then the small singular values carry information about the regions with little variation and can perhaps be omitted, whereas the large singular values carry information about more “dynamic” regions of the image

Suppose  $A$  has  $r$  nonzero singular values. For  $k \leq r$ , let

$$A_k = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \cdots + \sigma_k \mathbf{u}_k \mathbf{v}_k^T$$

For  $k = r$  we get the usual outer product form (2)

Load the image using `bmp::read.bmp`

```
my_image = bmp::read.bmp("../CODE/Julien_and_friend_1000x800.bmp")
my_image_g = pixmap::pixmapGrey(my_image)
my_image_g

## Pixmap image
##   Type           : pixmapGrey
##   Size           : 800x1000
##   Resolution     : 1x1
##   Bounding box   : 0 0 1000 800
```



## Doing the computations “by hand”

```
M = my_image_g@grey
MTM = t(M) %*% M
# Ensure matrix is symmetric
MTM = (MTM+t(MTM))/2
ev = eigen(MTM)
```

Given the size and nature of the entries, the matrix  $M^T M$  is symmetric only to  $1e-5$  precision, so we use a little trick to make it symmetric no matter what: take the average of  $M^T M$  and its transpose  $MM^T$

## Which version of the algorithm to use?

Make zero the eigenvalues that are close to zero (200 out of 1000)

```
ev$values = ev$values*(ev$values>1e-10)
```

Can we use the algorithm for all eigenvalues being distinct or do we have repeated ones?

```
any(duplicated(ev$values[ev$values>1e-10]))
```

```
## [1] FALSE
```

So we can use the standard algorithm

# Computing the SVD

```
idx_positive_ev = which(ev$values>1e-10)
sv = sqrt(ev$values[idx_positive_ev])
```

# Computing the SVD

Then  $D = \text{diag}(\sigma_1, \dots, \sigma_r)$ ,  $V$  is the matrix of normalised eigenvectors in the same order as the  $\sigma_i$  and for  $i = 1, \dots, r$

$$\mathbf{u}_i = \frac{1}{\sigma_i} A \mathbf{v}_i$$

ensuring that  $\|\mathbf{u}_i\| = 1$

```
D = diag(sv)
V = ev$eigenvectors[idx_positive_ev, idx_positive_ev]
c1 = colSums(V)
for (i in 1:dim(V)[2]) {
  V[,i] = V[,i]/c1[i]
}
```

# Computing the SVD

Finally, we compute the  $\mathbf{u}_i$ 's

```
U = M %*% V %*% diag(1/sv)

## Error in 'M %*% V':
## ! non-conformable arguments

r = length(sv)
im = list(u=U, d=sv, v=V)

## Error:
## ! object 'U' not found
```

## Using built-in functions

We can also use the built-in function `svd` to compute the SVD of  $M$

```
M.svd = svd(M)
```

The results are stored in a list with components  $u$ ,  $d$  and  $v$

## Make function to recreate an image from the SVD

Given the SVD `im` of an image and a number of singular values to keep `n`, we can recreate the image using the function `compress_image`

We output the new image, but also, the amount of information required to encode this new image, as a percentage of the original image size

```

compress_image = function(im, n) {
  if (n > length(im$d)) {
    # Check that we gave a value of n within range, otherwise
    # just set to the max
    n = length(im$d)
  }
  d_tmp = im$d[1:n]
  u_tmp = im$u[,1:n]
  v_tmp = im$v[,1:n]
  # We store the results in a list (so we can return other information)
  out = list()
  # First, compute the resulting image
  out$img = mat.or.vec(nr = dim(im$u)[1], nc = dim(im$v)[1])
  for (i in 1:n) {
    out$img = out$img + d_tmp[i] * u_tmp[,i] %*% t(v_tmp[,i])
  }
}

```

```

# Values of the "colours" must be between 0 and 1, so we shift and rescale
if (min(min(out$img)) < 0 ) {
  out$img = out$img - min(min(out$img))
}
out$img = out$img / max(max(out$img))
# Store some information: number of points needed and percentage of the
out$nb_pixels_original = dim(im$u)[1] * dim(im$v)[2]
out$nb_pixels_compressed = length(d_tmp) + dim(u_tmp)[1]*dim(u_tmp)[2] +
out$pct_of_original = out$nb_pixels_compressed / out$nb_pixels_original
# Return the result
return(out)
}

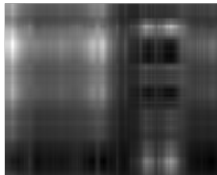
```

## Recreating the image

We can now recreate the image using the function `compress_image`

```
new_image = my_image_g
M.svd = svd(M)
M_tmp = compress_image(M.svd, 2)
new_image@grey = M_tmp$img
plot(new_image)
```

Using  $n = 2$  singular values



Uses 0.56% of the original information

Using  $n = 5$  singular values



Uses 1.41% of the original information

Using  $n = 10$  singular values



Uses 2.81% of the original information

Using  $n = 20$  singular values



Uses 5.63% of the original information

Using  $n = 50$  singular values



Uses 14.07% of the original information

# **The SVD (singular values decomposition)**

Singular values

The SVD

An application of the SVD – Image compression

**Computing the SVD**

Applications of the SVD – Least squares

Summary – Least squares methods

# Computing the SVD (case of $\neq$ eigenvalues)

To compute the SVD, we use the following result

## Theorem 77

*Let  $A \in \mathcal{M}_n$  symmetric,  $(\lambda_1, \mathbf{u}_1)$  and  $(\lambda_2, \mathbf{u}_2)$  be eigenpairs,  $\lambda_1 \neq \lambda_2$ . Then  $\mathbf{u}_1 \bullet \mathbf{u}_2 = 0$*

## Proof of Theorem 77

$A \in \mathcal{M}_n$  symmetric,  $(\lambda_1, \mathbf{u}_1)$  and  $(\lambda_2, \mathbf{u}_2)$  eigenpairs with  $\lambda_1 \neq \lambda_2$

$$\begin{aligned}\lambda_1(\mathbf{v}_1 \bullet \mathbf{v}_2) &= (\lambda_1 \mathbf{v}_1) \bullet \mathbf{v}_2 \\ &= A\mathbf{v}_1 \bullet \mathbf{v}_2 \\ &= (A\mathbf{v}_1)^T \mathbf{v}_2 \\ &= \mathbf{v}_1^T A^T \mathbf{v}_2 \\ &= \mathbf{v}_1^T (A\mathbf{v}_2) \quad [A \text{ symmetric so } A^T = A] \\ &= \mathbf{v}_1^T (\lambda_2 \mathbf{v}_2) \\ &= \lambda_2 (\mathbf{v}_1^T \mathbf{v}_2) \\ &= \lambda_2 (\mathbf{v}_1 \bullet \mathbf{v}_2)\end{aligned}$$

So  $(\lambda_1 - \lambda_2)(\mathbf{v}_1 \bullet \mathbf{v}_2) = 0$ . But  $\lambda_1 \neq \lambda_2$ , so  $\mathbf{v}_1 \bullet \mathbf{v}_2 = 0$



## Computing the SVD (case of $\neq$ eigenvalues)

If all eigenvalues of  $A^T A$  (or  $AA^T$ ) are distinct, we can use Theorem 77

1. Compute  $A^T A \in \mathcal{M}_n$
2. Compute eigenvalues  $\lambda_1, \dots, \lambda_n$  of  $A^T A$ ; order them as  $\lambda_1 > \dots > \lambda_n \geq 0$  ( $>$  not  $\geq$  since  $\neq$ )
3. Compute singular values  $\sigma_1 = \sqrt{\lambda_1}, \dots, \sigma_n = \sqrt{\lambda_n}$
4. Diagonal matrix  $D$  in  $\Sigma$  is either in  $\mathcal{M}_n$  (if  $\sigma_n > 0$ ) or in  $\mathcal{M}_{n-1}$  (if  $\sigma_n = 0$ )

5. Since eigenvalues are distinct, Theorem 77  $\implies$  eigenvectors are orthogonal set. Compute these eigenvectors in the same order as the eigenvalues
6. Normalise them and use them to make the matrix  $V$ , i.e.,  $V = [\mathbf{v}_1 \cdots \mathbf{v}_n]$
7. To find the  $\mathbf{u}_i$ , compute, for  $i = 1, \dots, r$ ,

$$\mathbf{u}_i = \frac{1}{\sigma_i} A \mathbf{v}_i$$

and ensure that  $\|\mathbf{u}_i\| = 1$

## Computing the SVD (case where some eigenvalues are $=$ )

1. Compute  $A^T A \in \mathcal{M}_n$
2. Compute eigenvalues  $\lambda_1, \dots, \lambda_n$  of  $A^T A$ ; order them as  $\lambda_1 \geq \dots \geq \lambda_n \geq 0$
3. Compute singular values  $\sigma_1 = \sqrt{\lambda_1}, \dots, \sigma_n = \sqrt{\lambda_n}$ , with  $r \leq n$  the index of the last positive singular value
4. For eigenvalues that are distinct, proceed as before
5. For eigenvalues with multiplicity  $> 1$ , we need to ensure that the resulting eigenvectors are LI *and* orthogonal

## Dealing with eigenvalues with multiplicity $> 1$

When an eigenvalue has (algebraic) multiplicity  $> 1$ , e.g., characteristic polynomial contains a factor like  $(\lambda - 2)^2$ , things can become a little bit more complicated

The proper way to deal with this involves the so-called Jordan Normal Form (another matrix decomposition)

In short: not all square matrices are diagonalisable, but all square matrices admit a JNF

Sometimes, we can find several LI eigenvectors associated to the same eigenvalue. Check this. If not, need to use the following

### Definition 78 (Generalised eigenvectors)

The vector  $\mathbf{x} \neq \mathbf{0}$  is a **generalized eigenvector** of rank  $m$  of  $A \in \mathcal{M}_n$  corresponding to eigenvalue  $\lambda$  if

$$(A - \lambda \mathbb{I})^m \mathbf{x} = \mathbf{0}$$

but

$$(A - \lambda \mathbb{I})^{m-1} \mathbf{x} \neq \mathbf{0}$$

## Procedure for generalised eigenvectors

$A \in \mathcal{M}_n$  and assume  $\lambda$  eigenvalue with algebraic multiplicity  $k$

Find  $\mathbf{v}_1$ , "classic" eigenvector, i.e.,  $\mathbf{v}_1 \neq \mathbf{0}$  s.t.  $(A - \lambda\mathbb{I})\mathbf{v}_1 = \mathbf{0}$

Find generalised eigenvector  $\mathbf{v}_2$  of rank 2 by solving for  $\mathbf{v}_2 \neq \mathbf{0}$ ,

$$(A - \lambda\mathbb{I})\mathbf{v}_2 = \mathbf{v}_1$$

...

Find generalised eigenvector  $\mathbf{v}_k$  of rank  $k$  by solving for  $\mathbf{v}_k \neq \mathbf{0}$ ,

$$(A - \lambda\mathbb{I})\mathbf{v}_k = \mathbf{v}_{k-1}$$

Then  $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$  LI

## Back to the normal procedure

With the LI eigenvectors  $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$  corresponding to  $\lambda$

Apply Gram-Schmidt to get orthogonal set

For all eigenvalues with multiplicity  $> 1$ , check that you either have LI eigenvectors or do what we just did

When you are done, be back on your merry way to step 6 in the case where eigenvalues are all  $\neq$

I am caricaturing a little here: there can be cases that do not work exactly like this, but this is general enough..

# **The SVD (singular values decomposition)**

Singular values

The SVD

An application of the SVD – Image compression

Computing the SVD

**Applications of the SVD – Least squares**

Summary – Least squares methods

# Pseudoinverse of a matrix

## Definition 79 (Pseudoinverse)

$A = U\Sigma V^T$  an SVD for  $A \in \mathcal{M}_{mn}$ , where

$$\Sigma = \begin{pmatrix} D & 0 \\ 0 & 0 \end{pmatrix}, \text{ with } D = \text{diag}(\sigma_1, \dots, \sigma_r)$$

( $D$  contains the nonzero singular values of  $A$  ordered as usual)

The **pseudoinverse** (or **Moore-Penrose inverse**) of  $A$  is  $A^+ \in \mathcal{M}_{nm}$  given by

$$A^+ = V\Sigma^+ U^T$$

with

$$\Sigma^+ = \begin{pmatrix} D^{-1} & 0 \\ 0 & 0 \end{pmatrix} \in \mathcal{M}_{nm}$$

# Least squares revisited

## Theorem 80

*Let  $A \in \mathcal{M}_{mn}$ ,  $\mathbf{x} \in \mathbb{R}^n$  and  $\mathbf{b} \in \mathbb{R}^m$ . The least squares problem  $A\mathbf{x} = \mathbf{b}$  has a unique least squares solution  $\tilde{\mathbf{x}}$  of minimal length (closest to the origin) given by*

$$\tilde{\mathbf{x}} = A^+ \mathbf{b}$$

*where  $A^+$  is the pseudoinverse of  $A$*

# **The SVD (singular values decomposition)**

Singular values

The SVD

An application of the SVD – Image compression

Computing the SVD

Applications of the SVD – Least squares

**Summary – Least squares methods**

# The least squares problem

## Problem Statement:

Given a system  $A\mathbf{x} = \mathbf{b}$  where  $A \in \mathcal{M}_{mn}$ ,  $\mathbf{x} \in \mathbb{R}^n$ ,  $\mathbf{b} \in \mathbb{R}^m$  (typically  $m > n$ ), find  $\tilde{\mathbf{x}}$  that minimizes

$$\|\mathbf{b} - A\mathbf{x}\|^2 = \sum_{i=1}^m (b_i - \sum_{j=1}^n A_{ij}x_j)^2$$

**Geometric interpretation:** Find the vector  $A\tilde{\mathbf{x}}$  in the column space of  $A$  that is closest to  $\mathbf{b}$

**Solution:**  $A\tilde{\mathbf{x}} = \text{proj}_{\text{col}(A)}(\mathbf{b})$

# Method 1: Normal equations

**The normal equations:**

$$A^T A \tilde{\mathbf{x}} = A^T \mathbf{b}$$

**When this works:**

- ▶ Always has at least one solution
- ▶ Any solution  $\tilde{\mathbf{x}}$  to the normal equations is a least squares solution

**Computational issues:**

- ▶ Forming  $A^T A$  can be numerically unstable
- ▶ Condition number of  $A^T A$  is the square of the condition number of  $A$
- ▶ Still useful for theoretical analysis

## Method 2: when $A$ Has linearly independent columns

**Condition:**  $A \in \mathcal{M}_{mn}$  has linearly independent columns

**Then:**  $A^T A$  is invertible and the least squares solution is **unique**

$$\tilde{\mathbf{x}} = (A^T A)^{-1} A^T \mathbf{b}$$

### Properties:

- ▶  $A^T A \in \mathcal{M}_n$  is square, symmetric, and positive definite
- ▶  $(A^T A)^{-1} A^T$  is called the *left pseudoinverse* of  $A$
- ▶ This gives the unique least squares solution

**Drawback:** Computing  $(A^T A)^{-1}$  directly can be numerically unstable

## Method 3: QR factorization

**QR Factorization:** If  $A \in \mathcal{M}_{mn}$  has linearly independent columns, then

$$A = QR$$

where  $Q \in \mathcal{M}_{mn}$  has orthonormal columns and  $R \in \mathcal{M}_n$  is upper triangular and nonsingular

**Least squares solution:**

$$\tilde{\mathbf{x}} = R^{-1} Q^T \mathbf{b}$$

**Advantages:**

- ▶ More numerically stable than forming  $A^T A$
- ▶  $R$  is upper triangular  $\Rightarrow$  solving  $R\tilde{\mathbf{x}} = Q^T \mathbf{b}$  by back substitution
- ▶ Condition number of  $R$  equals condition number of  $A$
- ▶ Gram-Schmidt or Householder reflections can compute QR factorization

## Method 4: Singular Value Decomposition (SVD)

**SVD:** For any  $A \in \mathcal{M}_{mn}$ ,

$$A = U\Sigma V^T$$

where  $U \in \mathcal{M}_m$  orthogonal,  $V \in \mathcal{M}_n$  orthogonal,  $\Sigma \in \mathcal{M}_{mn}$  with  $\Sigma_{ij} = \sigma_i \geq 0$  (singular values)

**Pseudoinverse:**  $A^+ = V\Sigma^+U^T$  where  $\Sigma^+$  has  $(\Sigma^+)_{ij} = 1/\sigma_i$  if  $\sigma_i > 0$ , else 0

**Least squares solution:**

$$\tilde{\mathbf{x}} = A^+ \mathbf{b}$$

**Key advantages:**

- ▶ Works for *any* matrix  $A$  (even when columns are linearly dependent)
- ▶ Gives the solution of *minimal length* when multiple solutions exist
- ▶ Most numerically stable method
- ▶ Reveals the rank of  $A$  through the number of non-zero singular values

## When to use which method

| Method             | When to use              | Advantages/Drawbacks           |
|--------------------|--------------------------|--------------------------------|
| Normal equations   | Theory, small problems   | Simple, but unstable           |
| $(A^T A)^{-1} A^T$ | $A$ has LI columns       | Explicit formula, unstable     |
| QR Factorization   | $A$ has LI columns       | Stable, efficient              |
| SVD                | Any $A$ , rank-deficient | Most stable, handles all cases |

Use QR for well-conditioned problems with LI columns, SVD for rank-deficient or ill-conditioned problems